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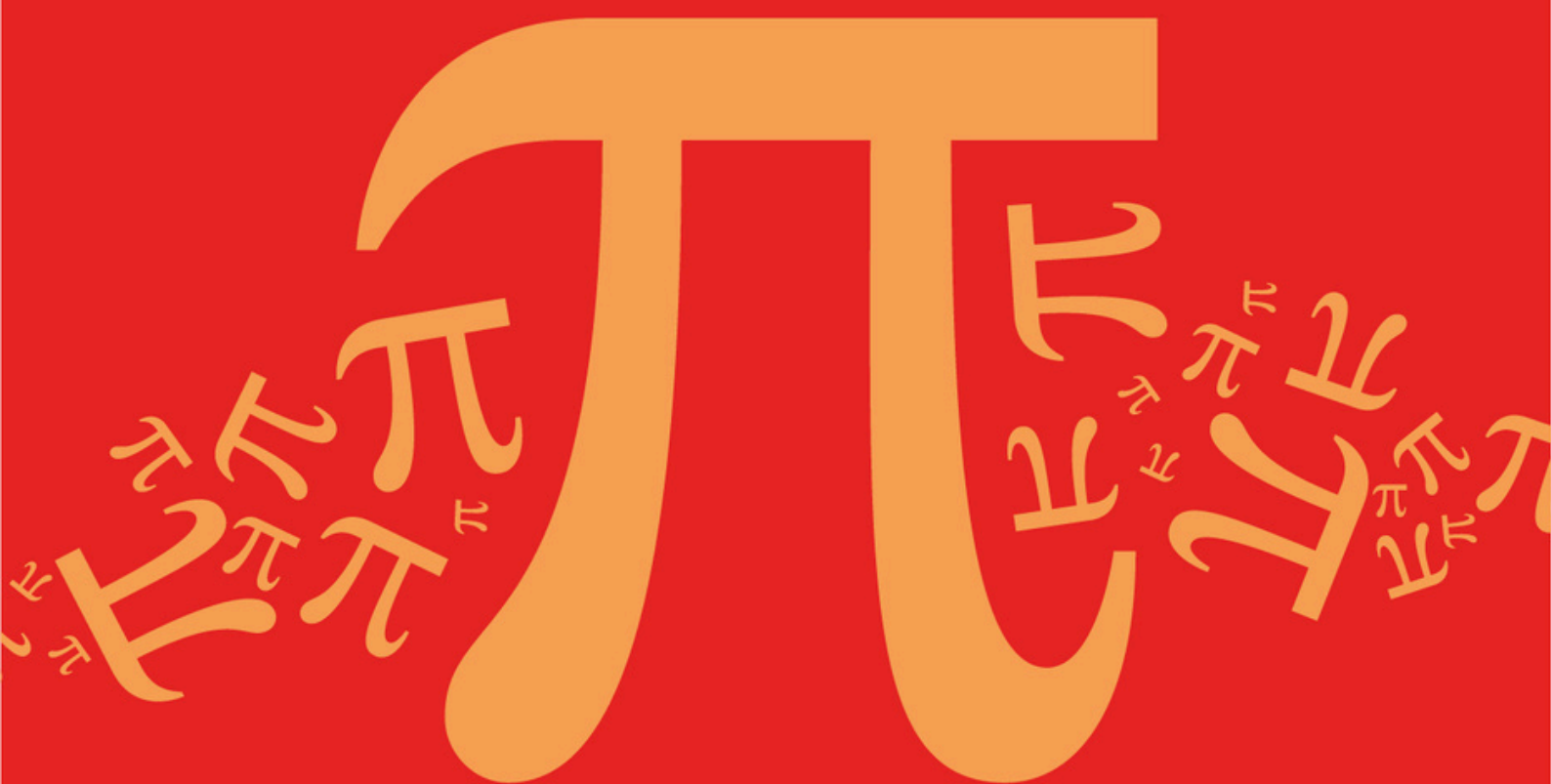
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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C4: Invariant Points and Lines

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

C: Matrices

Sub topic

C4: Invariant Points and Lines

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

Find two invariant points under the transformation given by $\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$

(Total 2 marks)

Q2.

The transformation T is defined by the matrix $\mathbf{M}$. The transformation S is defined by the matrix $\mathbf{M}^{-1}$. Given that the point (x, y) is invariant under transformation T , prove that (x, y) is also an invariant point under transformation S .

(Total 3 marks)

Q3.

The linear transformations T_1 and T_2 are represented by the matrices

$$\mathbf{M}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \quad \text{and} \quad \mathbf{M}_2 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{\sqrt{3}}{2} \\ 0 & 1 & 0 \\ -\frac{\sqrt{3}}{2} & 0 & \frac{1}{2} \end{bmatrix}$$

respectively.

(a) Give a full geometrical description of the transformations:

(i) T_1 ;

(2)

(ii) T_2 .

(3)

(b) Find the matrix which represents the transformation T_1 followed by T_2 .

(2)

(c) The linear transformation T_3 is represented by the matrix

$$\mathbf{M}_3 = \begin{bmatrix} k & 2 & -1 \\ 1 & 1 & 1 \\ 3 & 4 & 1 \end{bmatrix}$$

where k is a constant.

For one particular value of k , T_3 has a line L of invariant points.

(i) Find k .

(ii) Find the Cartesian equations of L in the form $\frac{x}{p} = \frac{y}{q} = \frac{z}{r}$.

(7)

(Total 14 marks)

Q4.

The plane transformation T is defined by

$$T : \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) A shape has an area of 3 square units. Find the area of the shape after being transformed by T .

(2)

- (b) (i) Find the equations of all the invariant lines of T .

(5)

- (ii) State the equation of the line of invariant points of T .

(1)

(Total 8 marks)

Q5.

- (a) Find the eigenvalues and corresponding eigenvectors of the matrix

$$M = \begin{bmatrix} 4 & 5 \\ 5 & 4 \end{bmatrix}.$$

(6)

- (b) The plane transformation T is given by the matrix M . Write down the coordinates of the invariant point of T .

(1)

(Total 7 marks)

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Q6.

Let $Y = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}.$

- (a) Show that 4 is a repeated eigenvalue of Y , and find the other eigenvalue of Y .

(7)

- (b) For each eigenvalue of Y , find a full set of eigenvectors.

(5)

- (c) The matrix Y represents the transformation T .

Describe the geometrical significance of the eigenvectors of Y in relation to T .

(3)

(Total 15 marks)

Q7.

$$\Delta = \begin{vmatrix} 16-q & 5 & 7 \\ -12 & -1-q & -7 \\ 6 & 6 & 10-q \end{vmatrix}$$

(a) It is given that

(i) By using row operations on the first two rows of Δ , show that $(4 - q)$ is a factor of Δ .

(2)

(ii) Express Δ as the product of three linear factors.

(4)

(b) It is given that $\mathbf{M} = \begin{bmatrix} 16 & 5 & 7 \\ -12 & -1 & -7 \\ 6 & 6 & 10 \end{bmatrix}$.

(i) Verify that $\begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix}$ is an eigenvector of $\mathbf{M}$ and state its corresponding eigenvalue.

(3)

(ii) For each of the other two eigenvalues of $\mathbf{M}$, find a corresponding eigenvector.

(7)

(c) The transformation T has matrix $\mathbf{M}$. Write down cartesian equations for any one of the invariant lines of T .

(2)

(Total 18 marks)

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Q8.

The transformation T is represented by the matrix $\mathbf{M}$ with diagonalised form

$$\mathbf{M} = \mathbf{U} \mathbf{D} \mathbf{U}^{-1}$$

where $\mathbf{U} = \begin{bmatrix} 4 & -1 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{D} = \begin{bmatrix} 27 & 0 \\ 0 & 1 \end{bmatrix}$.

(a) (i) State the eigenvalues, and corresponding eigenvectors, of $\mathbf{M}$.

(4)

(ii) Find a cartesian equation for the line of invariant points of T .

(2)

(b) Write down $\mathbf{U}^{-1}$, and hence find the matrix $\mathbf{M}$ in the form

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

where a , b , c and d are integers.

(5)

- (c) By finding the element in the first row, first column position of $\mathbf{M}^n$, prove that

$$4 - 3^{3n+1} + 1$$

is a multiple of 13 for all positive integers n .

(5)

(Total 16 marks)

Q9.

The plane transformation T is defined by

$$T: \begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{M} \begin{bmatrix} x \\ y \end{bmatrix}$$

where $\mathbf{M} = \begin{bmatrix} -1 & 4 \\ -1 & 3 \end{bmatrix}$.

- (a) Evaluate $\det \mathbf{M}$ and state the significance of this answer in relation to T .

(2)

- (b) Find the single eigenvalue of $\mathbf{M}$ and a corresponding eigenvector. Describe the geometrical significance of these answers in relation to T .

(5)

- (c) Show that the image of the line $y = \frac{1}{2}x + k$ under T is $y' = \frac{1}{2}x' + k$.

(3)

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(Total 10 marks)

Q10.

The non-singular matrix $\mathbf{M} = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}$.

- (a) (i) Show that

$$\mathbf{M}^2 + 2\mathbf{I} = k\mathbf{M}$$

for some integer k to be determined.

(3)

- (ii) By multiplying the equation in part (a)(i) by $\mathbf{M}^{-1}$, show that

$$\mathbf{M}^{-1} = a\mathbf{M} + b\mathbf{I}$$

for constants a and b to be found.

(3)

- (b) (i) Determine the characteristic equation of $\mathbf{M}$ and show that $\mathbf{M}$ has a repeated eigenvalue, 1, and another eigenvalue, 2.

(6)

- (ii) Give a full set of eigenvectors for each of these eigenvalues.

(5)

- (iii) State the geometrical significance of each set of eigenvectors in relation to the transformation with matrix $\mathbf{M}$.

(3)

(Total 20 marks)

Q11.

A plane transformation is represented by the 2×2 matrix $\mathbf{M}$. The eigenvalues of $\mathbf{M}$ are 1 and 2,

with corresponding eigenvectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ respectively.

- (a) State the equations of the invariant lines of the transformation and explain which of these is also a line of invariant points.

(3)

- (b) The diagonalised form of $\mathbf{M}$ is $\mathbf{M} = \mathbf{U} \mathbf{D} \mathbf{U}^{-1}$, where $\mathbf{D}$ is a diagonal matrix.

- (i) Write down a suitable matrix $\mathbf{D}$ and the corresponding matrix $\mathbf{U}$.

(2)

- (ii) Hence determine $\mathbf{M}$.

(4)

- (iii) Show that $\mathbf{M}^n = \begin{bmatrix} 1 & f(n) - 1 \\ 0 & f(n) \end{bmatrix}$ for all positive integers n , where $f(n)$ is a function of n to be determined.

(3)

(Total 12 marks)

Q12.

The transformation T maps (x, y) to (x', y') where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) Describe the difference between an invariant line and a line of invariant points of T .

(1)

- (b) Evaluate the determinant of the matrix $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and describe the geometrical significance of the result in relation to T .

(2)

- (c) Show that T has a line of invariant points, and find a cartesian equation for this line. (2)
- (d) (i) Find the image of the point $(x, -x + c)$ under T . (2)
- (ii) Hence show that all lines of the form $y = -x + c$, where c is an arbitrary constant, are invariant lines of T . (2)
- (e) Describe the transformation T geometrically. (3)
- (Total 12 marks)

Q13.

A transformation is represented by the matrix $\mathbf{A} = \begin{bmatrix} 0.28 & -0.96 & 0 \\ 0.96 & 0.28 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

- (a) Evaluate $\det \mathbf{A}$. (1)
- (b) State the invariant line of the transformation. (1)
- (c) Give a full geometrical description of this transformation. (3)
- (Total 5 marks)

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Q14.

The transformation T is a stretch in a fixed direction, and maps the point (x, y) to the image point (x', y') , where

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 6.4 & -7.2 \\ -7.2 & 10.6 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

- (a) Determine the scale factor of the stretch. (2)
- (b) By considering the invariant points of T , or otherwise, find an equation for the line of invariant points of T , giving your answer in the form $y = mx$. (5)
- (Total 7 marks)